

Lab 1: Use-Case Flow Specification

Problem Statement #23: Hyperlocal Courier Dispatch & Tracking Engine

Use Case: UC-01 – Create Courier Request

Primary Actor: Sender Client

Supporting Actors: Delivery Rider, External Map / GPS Service

Goal: The sender creates a courier request and the system assigns it to the nearest active delivery rider within a 3 km radius.

Preconditions

- The Sender Client can access the courier platform.
- Pickup, delivery and parcel details are available.
- The system has current rider location and availability data.

Postconditions

- The courier request is created with a request ID.
- The nearest eligible rider within 3 km receives the dispatch request.
- The Sender Client can view the request status after dispatch.

Main Success Scenario

1. The Sender Client selects the option to create a courier request.
2. The system asks for the pickup location, delivery location and parcel details.
3. The Sender Client enters the required details and submits the request.
4. The system validates the details and creates the courier request.
5. The system checks for active delivery riders within 3 km of the pickup location.
6. The system selects the nearest eligible rider and sends the rider a dispatch request.
7. The rider accepts the request.
8. The system changes the request status to Accepted and notifies the Sender Client.
9. The use case ends successfully.

Alternate Flow - A1: No Active Rider Within 3 km

- A1.1 At Step 5, the system finds no active rider within the 3 km radius.
- A1.2 The system does not assign the request to an offline or out-of-range rider.
- A1.3 The Sender Client is informed that no nearby rider is currently available.
- A1.4 The request remains unassigned and can be retried when a rider becomes available.